<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks Available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>8</td><td>(a)</td><td></td><td></td><td>
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Parents in 1st Punnett square (1)
Parents in 2nd Punnett square (1)
Both crosses (ecf) (1)
chance: 25 % compared with 50% (ecf) (1) – conclusion must be consistent with Punnett squares</td><td></td><td>4</td><td></td><td>4</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>White blood cells / leucocyte / lymphocyte / phagocyte / macrophage (1)

+ any 2 × (1) from:
ingesting bacteria (1)
producing antibodies (1)
producing antitoxins (1)</td><td>3</td><td></td><td></td><td>3</td><td></td><td></td></tr>
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Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			any 3 × (1) from: Short term effects: breathing rate increases (1) to provide the oxygen and remove carbon dioxide faster (1) Long term effects: the body becomes more efficient at transporting oxygen (1) Increased lung capacity (1) Accept: breathing rate decreases (1) Accept: clears mucus more efficiently (1)	3			3		
	(d)	(i)		A = 0.5 (1) B= 0.83 (1) C = 0.87/0.8666……. (1) C>B>A / Volume higher in CF and higher again in CF with infection (1)	1	3		4	3	
		(ii)		The greater the breaths per minute (1) the lower the blood oxygen level (1) accept comparison with volume of air per breath for (1) only		2		2		
				Question 8 total	**7**	**9**	**0**	**16**	**4**	**0**